Detecting Seismic Activity with a Covariance Matrix Analysis of Data Recorded on Seismic Arrays

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Abstract

Modern seismic networks are recording the ground motion continuously all around the word, with very broadband and high-sensitivity sensors. The aim of our study is to apply statistical array-based approaches to processing of these records. We use the methods mainly brought from the random matrix theory in order to give a statistical description of seismic wavefields recorded at the Earth's surface.

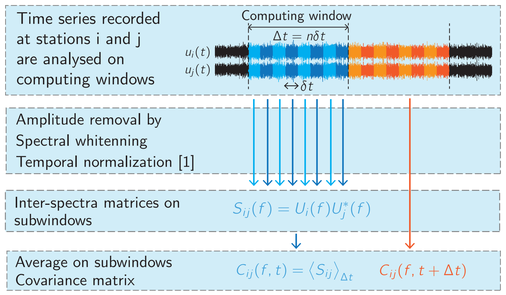
We here present how we estimate the array covariance matrix and explore the distribution of its eigenvalues that contains information about the coherency of the sources that generated the studied wavefields.

We investigate the detection capacity of our methods at different scales and in different frequency ranges by applying it to the records of two networks: (1) the seismic monitoring network operating on the Piton de la Fournaise volcano at La Réunion island composed of 21 receivers and with an aperture of ~15 km, and (2) the transportable component of the USArray composed of ~400 receivers with ~70 km inter-station spacing.

Method : covariance matrix analysis

Array covariance matrix $C_{ij}(f,t) = \left\langle U_i(f) U_j^*(f) \right\rangle_{\Delta t}$

Computational scheme



Main parameters

- Pre-processing is performed in order to normalize energy,
- Subwindows length δt is chosen as small as possible (constrained by Fourier transform and waves time travel)
- The number of subwindows n is chosen as large as possible to ensure average relevance, while keeping Δt as short as possible

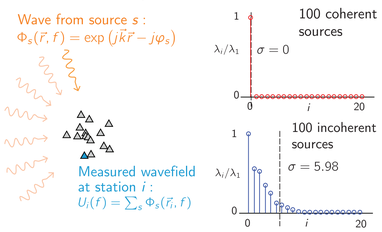
Covariance matrix spectral width

Eigenvalue decomposition of C : (λ_i are eigenvalues and $\mathbf{v}_i$ eigenvectors) $\mathbf{C} = \sum_i \mathbf{v}_i^\dagger \lambda_i \mathbf{v}_i$

Covariance matrix spectral width : (using descending ordered eigenvalues) $\sigma = \frac{\sum_i i\lambda_i}{\sum_i \lambda_i}$

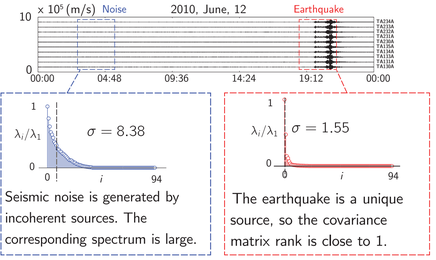
Synthetic tests

Covariance matrix analysis as a function of the sources cohenrency



Example using US Transportable array

Comparison between covariance matrix analysis on micro-seismic noise and the Nicobar Island earthquake (M=7.5)

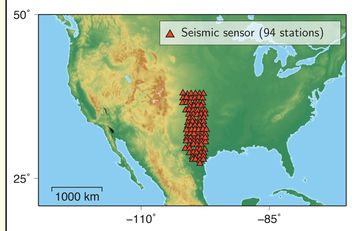


Seismic noise is generated by incoherent sources. The corresponding spectrum is large.

The earthquake is a unique source, so the covariance matrix rank is close to 1.

Application to USArray

Data



- Vertical component analysis
- 94 stations from US transportable array
- Aperture : 600km
- Interstation distance : 70km

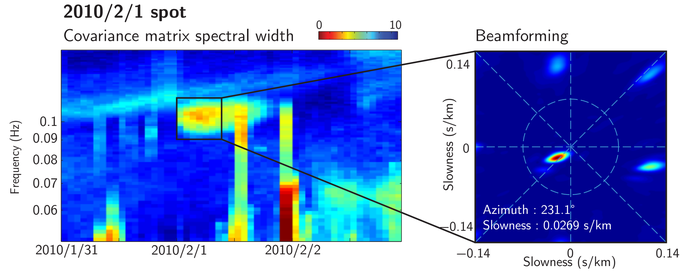
Beamforming from covariance matrix

$$\mathrm{Beam}(\gamma,\theta) = \sum_{i=1}^{N}\sum_{j=1}^{N} C_{ij}\phi_i(\gamma,\theta)\phi_j^*(\gamma,\theta)$$

$$\phi_i(\gamma,\theta) = e^{2\pi j\gamma f(x_i cos(\theta) + y_i sin(\theta)}$$

Beamforming can directly be computed from covariance matrix using phase shifting test functions. We assume that the incoming waves are plane, propagating in the network plane. The obtained result thus indicates the projection of the projected wave slowness on the network plane.

2010/2/1 spot



A rise of coherent energy between 0.09 and 0.11 Hz is clearly visible all during the day 2010/2/1 on the covariance spectral width.

We apply beamforming analysis on the zone delimited by the square. The azimuth of the beamformer indicates that the energy of the spot comes from the south-west (231°) with a apparent slowness around 0.027 s/m which indicates that the wave is volumic.

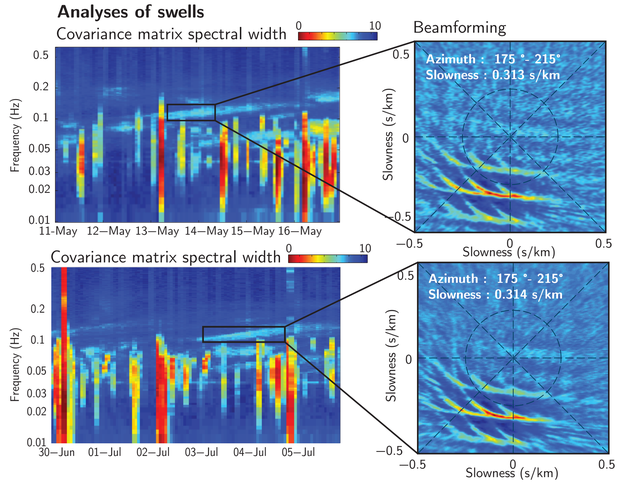
If you want to know more about geolocalization of this spot, please feel free to see the poster of Lise Retailleau *et. al* :

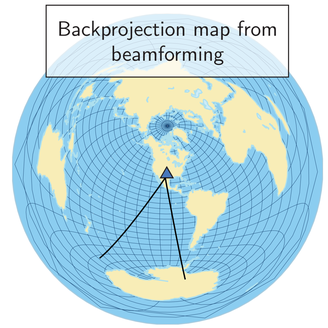
"Automatic Detection and Location of Teleseismic Events with Using Usarray As a Big Antenna"

Poster reference: 4514

Session: General Contributions in Seismology (S43A)

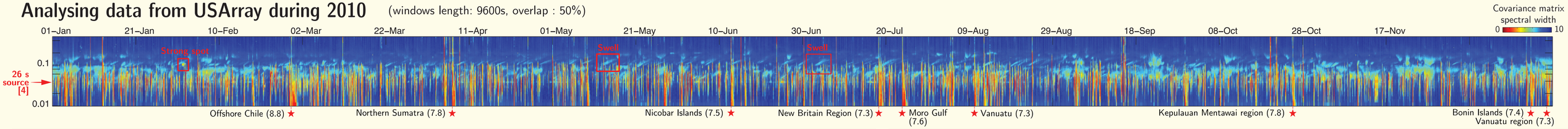
Analyses of swells





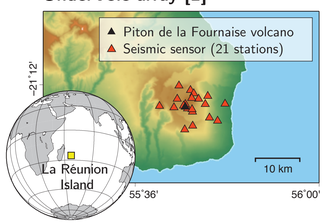
Analysing data from USArray during 2010

(windows length: 9600s, overlap : 50%)



Application to Piton de la Fournaise volcano, La Réunion Island

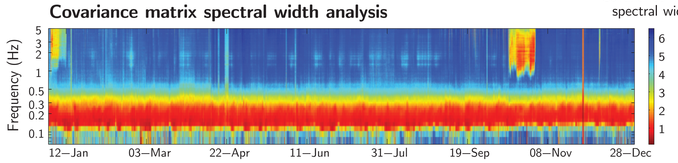
UnderVolc array [2]



Data

- Continuous records from 2010/1/1 to 2010/12/31
- Vertical component
- Array aperture : ~ 10 km
- Interstation distance : ~ 3 km

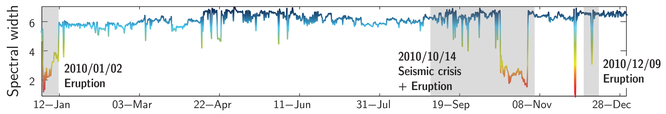
Covariance matrix spectral width analysis



Windows length : 2700 s (45 minutes)

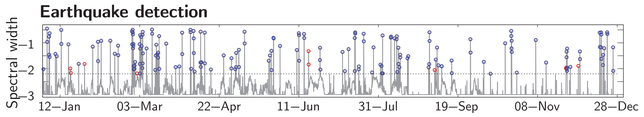
Subwindows length : 27 s

Detecting the Piton de la Fournaise volcanic activity



By averaging the covariance matrix spectral width between 1 and 5 Hz, we can visualize the volcanic activity. We compare the results with a description of the volcano given in [3].

Earthquake detection

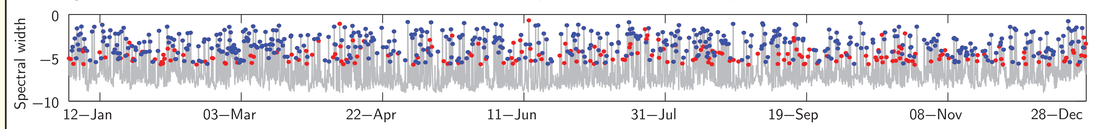


- 190 total detection
- 11 false detection
- 67 non detected global M>6 and local M>1 earthquakes during 2010

Detecting earthquakes during 2010 using USarray

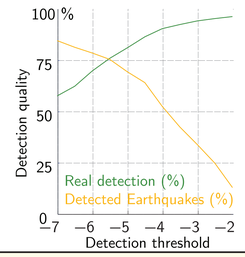
Detecting teleseismic activity for earthquakes of M > 5.2

Average between 0.01 and 0.08 Hz and threshold detection and Comparision with USGS earthquake database



- 701 good detections over 928 detections (75%)
- 723 corresponding M > 5.2 earthquakes over 928 earquakes occurance during the period (76%)

Trade-off the real detection maximization and the number of detected earthquakes mainly depends on the detection threshold.



References

[1] Bensen, G. D., et al. "Processing seismic ambient noise data to obtain reliable broad-band surface wave dispersion measurements." Geophysical Journal International 169.3 (2007): 1239-1260.

[2] Brenguier, Florent, et al. "From high quality seismic data acquisition in remote volcanic area to fast data distribution to scientific community: The UnderVolc project on Piton de la Fournaise volcano." EGU General Assembly Conference Abstracts. Vol. 12. 2010.

[3] Roult, Geneviève, et al. "A new comprehensive classification of the Piton de la Fournaise activity spanning the 1985–2010 period. Search and analysis of short-term precursors from a broad-band seismological station." Journal of Volcanology and Geothermal Research 241 (2012): 78-104.

[4] Shapiro, Nikolai M., M. H. Ritzwoller, and G. D. Bensen. "Source location of the 26 sec microseism from cross-correlations of ambient seismic noise." Geophysical research letters 33.18 (2006).